

My Word is My Bond App

(Sözüm Söz Uygulaması)

AI-Supported Digital Commitment and Well-Being Application

Team: Group 8

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Generative AI Models: gpt-4o-mini (OpenAI-compatible) and Gemini-3.6-flash (Google Gemini)

AI Integration: Direct browser API calls

Technology Stack: HTML5, CSS3, and Vanilla JavaScript

Problem and Solution

Digital platforms are now a central part of everyday life, but smartphone use can gradually shift from intentional behavior to automatic use. According to the Republic of Türkiye Ministry of Transport and Infrastructure, people in Türkiye spend an average of **25 hours and 4 minutes per week on social media**, while weekly online media use reaches **41 hours and 37 minutes**. These figures do not directly indicate digital addiction, but they demonstrate the importance of digital self-regulation.

Most screen-time tools tell users how much time they spent after the behavior had already occurred. **SÖZÜM SÖZ takes a preventive approach:** every time users open an application, they first decide how long they intend to stay and make a voluntary “**time promise.**”

For example: **Instagram → 20-minute promise**

The session is then monitored. Keeping the promise earns **+3 points**, while exceeding the selected duration or rapidly moving to another application results in **−5 points** and a non-judgmental awareness notification. The current prototype automatically penalizes rapid application transitions; a future version will allow users to confirm whether a transition was intentional.

The system also includes streaks, a daily promise goal, weekly challenges, a visual status emoji, daily and weekly AI reflections, graphs, an AI chatbot, and a reward catalogue. Rather than rewarding low screen time itself, **SÖZÜM SÖZ rewards consistency with the limits users choose for themselves**. Its central difference can be summarized as **Existing screen-**

time tools measure how long users stayed. SÖZÜM SÖZ compares how long users intended to stay with what actually happened.

Generative AI and Prototype

The prototype supports two selectable AI providers: an OpenAI-compatible chat-completions interface using **gpt-4o-mini by default and** Google Gemini using **gemini-3.6-flash** through *"generateContent."* Calls are made directly from the browser.

Generative AI is used for four main functions:

- **Daily Reflection:** Summarizes the user's own recent behavior without judgment.
- **AI Coach Notifications:** Provides contextual feedback and realistic micro-break suggestions after exceeded commitments.
- **Weekly Summary:** Identifies possible patterns and proposes a small behavioral experiment.
- **Multi-turn Chatbot:** Provides digital well-being support using anonymized usage context.

For example, instead of stating, *"You are addicted to social media,"* the AI may generate:

"You exceeded several of your evening commitments this week. This may indicate a recurring pattern. You could experiment with a shorter evening limit followed by a brief screen-free break."

Every generated item is visibly marked as **AI-generated** and ends with a notice explaining that the information is intended for awareness and **does not constitute medical diagnosis or treatment advice.**

The application is also **offline-first**. If no API key or internet connection is available, deterministic fallback responses following the same safety rules allow the prototype to continue working end-to-end. User data is stored locally using *"localStorage"*, while only anonymized aggregates are sent to an AI provider when live AI is enabled.

Prompt Design

Prompt engineering was deliberately treated as part of the product architecture. We combined several techniques covered in the module rather than relying on a single open-ended instruction.

The main techniques include:

-Zero-shot role prompting: The system defines the AI as a non-judgmental digital well-being coach and establishes fixed behavioral and medical-safety rules.

-Few-shot prompting: Example outputs are provided for daily reflections and coaching notifications to maintain a consistent tone across different models.

-Structured step-by-step prompting: Weekly summaries follow a fixed sequence such as **summary → pattern → possible trigger → small experiment → safety warning**.

Prompts also use clear delimiters separating instructions, user data, examples, and verification steps. Certain outputs are constrained to a predefined **JSON schema**, enabling reliable rendering in the application.

A self-check step asks the model to verify whether every numerical statement is supported by the supplied data and whether the response contains medical, judgmental, or unsupported claims. To bound context and cost, the model receives only the most recent 20 chat turns, while a short 40-turn window is kept on-device for display.

Additionally, crisis, medical, and off-topic inputs are processed through deterministic **guard layers before reaching the model**. The application includes a **“Show the Prompt Used”** function, so the prompt structure can be demonstrated directly during evaluation.

Key prompt (abridged) — full text in `src/js/ai.js`, also viewable in-app via "Show the Prompt Used":

"You are 'SÖZÜM SÖZ' ('my word is my bond'), a non-judgmental digital well-being coach. (1) Do not use medical or stigmatizing words ('addict', 'patient', 'treatment') and do not diagnose. (4) Use only numbers present in the supplied data — if unsure, say 'this is not in my records'. (7) End every output with: 'This information is for awareness purposes and is not a medical diagnosis or treatment recommendation.' (8) On a crisis signal (suicide, self-harm, hopelessness), without diagnosing, calmly direct to the Yeşilay 115 line and 112. (9) VERIFICATION: is every number I used present in the data? Does the response contain any medical, blaming, or unsupported wording?"

Rationale: fixing the role and rules in one system prompt is our strongest zero-shot guard; rules 4/9 prevent fabricated numbers, rule 1 prevents diagnosis and blame, rules 7/8 enforce transparency and crisis safety.

Ethical Evaluation and Privacy

Privacy and responsible AI use are particularly important because SÖZÜM SÖZ processes behavioral information.

The prototype begins with a full KVKK m.10 privacy notice and explicit consent screen, shown the first time the SÖZÜM SÖZ icon is tapped; without consent the application stays locked. Users can withdraw consent (KVKK m.7), download their stored data as JSON (KVKK m.11 right of access), or permanently erase all data including consent (KVKK m.11 right of erasure — the app then relocks and re-asks consent). Data is stored only in browser localStorage; when live AI is enabled, only anonymized behavioral aggregates are sent to the provider the user selects (KVKK m.8–9), never names, photographs, or contact information.

The main risks and safeguards are:

- **Medical overclaiming:** The AI is prohibited from diagnosing digital addiction or other medical conditions.
- **Hallucination:** Numerical claims must come only from supplied usage data.
- **Judgment and stigma:** Accusatory or clinical language is prohibited.
- **Privacy:** Unnecessary demographic and personal data are not collected.
- **Crisis situations:** Predefined guards provide calm, non-diagnostic guidance toward Yeşilay 115 and emergency 112 where appropriate.
- **AI transparency:** AI-generated content is clearly labelled.
- **Bias:** Personalization is based on the user's own recent behavior rather than demographic profiling.

A future response-filtering layer will provide an additional validation step before generated content reaches the user.

Gamification, Yeşilay, and Future Development

Users earn points by keeping their promises and maintaining streaks. Streak bonuses scale with consistency: 3/7/14/30 promises in a row → +10/+25/+60/+150. A daily promise goal rewards keeping 3 promises in one day (+20), and five optional weekly challenges (screen time < 10 h, 7 on-time closures, 5 micro-breaks, promises in 3 different apps, fewer than 3 overshoots) nudge behaviors such as taking micro-breaks and finishing sessions on time.

As a proposed future Yeşilay collaboration, accumulated points could be exchanged for products such as **children's books, piggy banks, bottles, thermoses, clocks, or other campaign products**. Keeping at least 30 promises in a month with no broken promises provides eligibility for a monthly draw (prizes: travel/holiday package, laptop, smartphone, game console, and a Yeşilay experience camp — presented as a scenario, not an existing sponsorship). This is presented as a **future partnership proposal, not an existing sponsorship agreement**.

A dedicated **SÖZÜM SÖZ Kids** version is also planned, with age-appropriate language, positive reinforcement, stronger privacy safeguards, and parental consent.

Because the hackathon period was limited, we prioritized a reliable working core. The prototype currently includes a fast demo mode where approximately **1 second represents 1 minute**, allowing realistic usage scenarios to be demonstrated quickly. The project has **87 automated tests with 0 failures and a 5/5 build check**, providing concrete evidence that the core functionality operates consistently.

Future improvements include real operating-system usage integration, intentional-transition confirmation, stronger AI response filtering, expanded personalized challenges, the Kids version, and a real Yeşilay-supported reward infrastructure.

SÖZÜM SÖZ does not aim to remove technology from people's lives. It aims to help people use technology on their own terms and keep the digital promises they make to themselves.

References

Republic of Türkiye Ministry of Transport and Infrastructure. (2026). *Haftada 2 Günümüz İnternette Geçiyor*.

Samsung Innovation Campus. (2026). *Generative AI Hackathon Guide – Digital Addiction and Digital Well-Being*.